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Write-Up

GitHub: https://github.com/alfalfa14/aigc-art-trends-dashboard

Render: https://aigc-art-trends-dashboard.onrender.com

The dataset used for the DIGS 20004 Final project was a dataset named *AI-Generated Art Trends Dataset*, found on Kaggle. It has a usability score of 10.00 and a license of Apache 2.0. It contains detailly recorded AI-generated artworks created between 2022 and 2024, and each record includes fields such as artwork ID, creation date, art style, medium, tools used, popularity score, region of origin, platform, and a brief description (Ali). The dataset was selected for the final project because of the wide range of details it covers about each piece of artwork recorded in the dataset. This makes it well-suited for a multi-dimensional dashboard for users to explore how AI art trends have evolved across regions and over time.

The three visualizations were designed to provide viewers with insights from different dimensions of the analysis of the dataset. The first chart is a line chart showing the percentage share of each art style per year. The motivation was to track which styles were increasing or decreasing the attention. Using percentage share rather than count in the line chart allows a more adequate comparison across years that may have different total numbers of artworks. A checklist control lets the viewer select and deselect styles they intend to analyze, making it easy to focus on specific comparisons. Hover interaction reveals the details for each year, allowing more information without making the map overly sophisticated. The second chart is a heatmap showing artwork count for each region and the corresponding art style. The motivation was to reveal whether different parts of the world have appreciations for different styles. A heatmap was chosen because it communicates a two-dimensional graph, while the darkness or brightness of the color represents the count vividly. Hover interaction reveals the details for each combination, allowing more information without making the map overly sophisticated. The third chart is a bubble map plotted on a world map. Each bubble represents one region, with size representing the number of artworks and color suggesting the average Popularity Score (as illustrated in *AI-Generated Art Trends Dataset*). The motivation was to give the dashboard and the viewers a geographic dimension and to show both volume and engagement in a single view. Hover interaction reveals the details for each region, allowing more information without making the map overly sophisticated.

Several key findings emerged from the visualizations. In the “Style Trend” chart, no single style dominated consistently across all three years. Most styles fluctuated within a rather narrow range of 9% to 11%, suggesting that the AI art aesthetic remains relatively diverse. The “Artwork Count by Region and Style” heatmap revealed that Europe produced the highest concentration of Cubism and Expressionism artworks compared to other regions, while North America showed a stronger preference toward Surrealism. On the other hand, Oceania had the lowest overall artwork count across a high proportion of styles, which might reflect lower platform activity of AI-generated Art in that region. The “World Map” confirmed that artwork volume is distributed quite evenly across all continents. Europe and Asia produced slightly more artworks overall. The average Popularity Scores are the highest in Europe and North America, indicating higher viewer engagement.

If given all the resources in the world, there are several directions to develop the project. First, the dataset can be specified by countries instead of only the continents. This allows a more precise analysis of the regional difference. For example, China and Saudi Arabia are on the same continent but have totally different cultures, so there is likely to be a great difference in the AI-generated art from these two countries. Second, the Popularity Score is quite vague for the viewers. There can be a breakdown graph showing what the score exactly means and how it is acquired from different platforms. Third, the dataset includes the tools used for each artwork. A detailed analysis of the tools might illustrate more insights about the artworks for the viewers.

Works Cited

Ali, Waqar. (2024). AI-Generated Art Trends Dataset [Data set]. Kaggle. https://www.kaggle.com/datasets/waqi786/ai-generated-art-trends